

FormPix Raspberry Pi Setup Guide

Complete Installation & Configuration of FormPix

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1. Requirements & Hardware

Setting up a Raspberry Pi for FormPix is super simple. Just use this as a guide when connecting a new Pi to FormPix

1.1 What You'll Need

- Raspberry Pi 4 Model B
- MicroSD card
- A machine capable of running SSH to the Pi
- 1 Ethernet cable
- 1 PSU
- 1 Power cord
- Jumper wires (like a couple I thinks)
- USB microSD card reader

2. Flashing the OS

Take the microSD card and plug it into the pc using the USB reader

2.1 Download Raspberry Pi Imager

Download the official Raspberry Pi Imager from this website

<https://www.raspberrypi.com/software/> and download the Raspberry Pi Imager for the OS you are imaging the device on

2.2 Writing the Image

1. Insert the MicroSD card into your computer
2. Open Raspberry Pi Imager and select Raspberry Pi 4
3. Choose Raspberry Pi OS (64-bit)
4. Select your MicroSD card as the storage target
5. For customization, name the pi what you want to have it be identified as when scanning (e.g., smifferBar)
6. Set the capital city and timezone the Pi should be in
7. Enter the desired user and password
8. Skip Wi-Fi, as you will be using Ethernet to connect to the Pi

9. Ensure SSH is enabled; you will not be able to connect to the Pi via terminal if this is disabled

10. Click Write and then click I understand, erase, and write

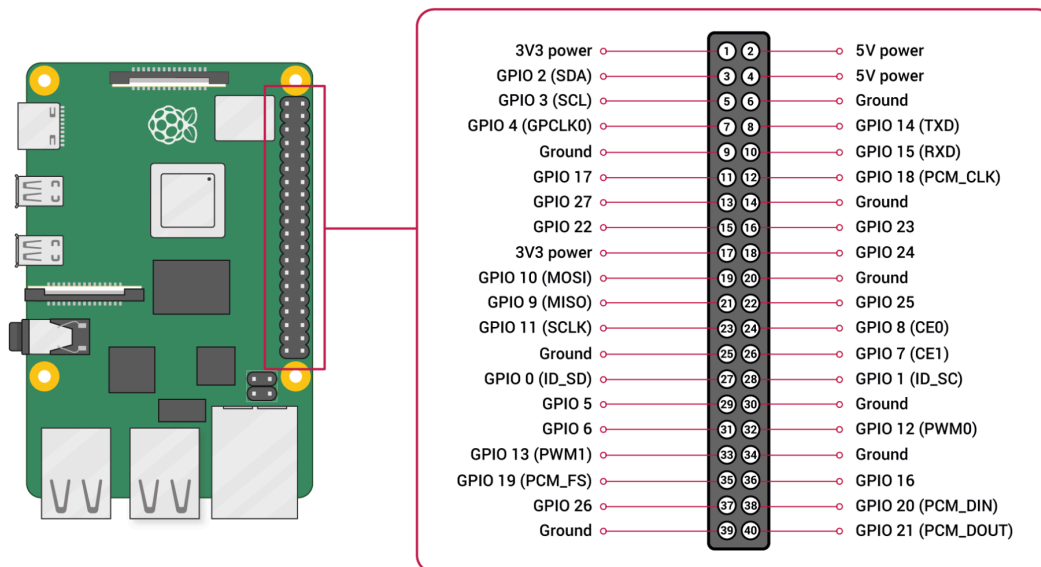
11. Safely eject the card when finished

3. Configuring the PSU and WS2812B neopixels

This is where things get technical and confusing. If you are at all confused by these instructions, then uh idk twin just figure it out. I guess it's not that hard, just make it workie if no workie then scream steven fix it (R.I.P Steven ? - 2026. He is with us in spirit), and tell smiffers it doesn't work.

3.1 Understanding GPIO pins

When working with Neopixels and PSU wiring, it's best if you understand GPIO pins as well. According to my sources ([Google.com](https://www.google.com)) "GPIO (General Purpose Input/Output) pins are uncommitted, programmable signal pins on microcontrollers (like Arduino) or computers (like Raspberry Pi) that act as the interface between the processor and external components" so those tiny little pins on the side of the Pi are the GPIO pins that is where you'll be connecting your lights and PSU. If at all you need to reference where to plug something in, refer to the image below for each pin label.



3.2 Connecting the Neopixels

Assuming you have brand-new lights, take the lights out of the package first, then unravel them. Look at the strip, you should notice 3 markings: **GND**, **DIN**, and **5V**. Take note of each wire that leads from each label. Take the **GND** wire and connect it to

physical pin 6 on the Pi, connect **DIN** to GPIO pin 21 (physical pin 40), and connect the **5V** to physical pin 2 for the power to the strip. Later on, if you find the light strip needing more voltage/power, there is an extra white wire and a red wire. These wires are **5V** (red wire) and **GRD** (white wire), you are able to connect these into the power supply for extra power to the lights.. That should be enough to at least test the lights. If you would like to do that, then you can skip step 3.3 for now and come back to it later.

3.3 Connecting the PSU to the Wall Cable

Lowkey, this is one of the most important parts of the whole setup. If you wire the PSU wrong, best case scenario nothing works. Worst case scenario, the wires become emotionally unstable and start producing fire (this has happened maybe once or twice before in the great year of 2026). Before touching anything, make absolutely sure the PSU is **NOT plugged into the wall**. Electricity is cool until it starts fighting back and boy touching you.

On the power supply, you'll see a bunch of labels and weird little symbols above the screw terminals. Usually there will be an **L**, an **N**, and the little ground symbol " $\equiv$ ". The **L** stands for Live, the **N** stands for Neutral, and the ground symbol is well ground you peanut brain. After you finish wiring everything, these terminals should be covered while the PSU is powered on so nobody accidentally finger-blasts 120 volts into your tiny little bodies. For these three there should be ONLY the wires from the one power cord and no other cable. Make sure to not let any wire boy touch other wires and cause a ka-boom. After finishing this whole step (3.3), you don't need to connect the pi with a USB-C cable anymore and only need to plug in the power cord of the power supply. If you find yourself needing the USB-C cable, something is wrong. Lowkey a cool idea for y'all to do is make a switch, on the pi case that Christopher Smith will 3D print out for you, that controls the power to the pi so you can just switch off the power on and off. If you do this, document it. Better off make it control the power from the power cord so the whole PSU won't be live.

The image below is a close representation of a power supply you will be using.



(As you see there are labels above the screw connections)

3.3.1 Connecting the wires to the PSU

Now grab the multimeter. I know you are programmers and seeing real hardware probably causes emotional distress, but you don't got this. Turn the dial to the resistance section, the one with the " Ω " symbol. Any range works honestly, but something like "2K" is perfectly fine.

Take the power cord you're using for the PSU. By this point I'm assuming you already stripped the outer insulation and exposed the wires inside. Put one multimeter probe on one of the plug pins, then touch the other probe to one of the stripped wires on the opposite end. If the multimeter shows a number REALLY close to 0 (basically 0), congratulations, those are connected. If it doesn't, then you are touching the wrong wire so just try the other wires.

Once you figure out which wire goes to which plug pin, mark them somehow so your goldfish-memory brain doesn't forget in 30 seconds. Then look at the PSU terminals and match everything up. The wire for **L** goes into the terminal labeled **L**. Crazy concept, I know. Same thing for **N** and ground. Tighten the screws down properly and double-check everything before plugging the PSU into the wall. Electricity does not believe in second chances.

3.4 Connecting the PSU to the Raspberry Pi

For this part, you need two male-to-female jumper wires from somewhere in the room. If Smith moved the IRL Tech supplies again, you have two choices. You can calmly ask where they are like a civilized human being, OR you can yell at full volume: "DARN STEVEN AND SMITH POOKIE BEAR!!!! WHERE DID YOU PUT THOSE SILLY WIRES!!!" Statistically speaking, the second method has a higher success rate.

Once you acquire the wires, you have two options for connecting them to the PSU. Option A is stripping the male ends and screwing the bare wire directly into the PSU terminals like a normal person. Option B is just shoving the male pins directly into the terminal openings like a goblin. Both methods work fine as long as the connection is secure and not wobbling around.

On the power supply, connect one wire to the **5V output** and the other wire to the **negative (-) or ground output**. These will eventually connect to the Raspberry Pi power pins. Make VERY sure you don't reverse them unless your long-term goal is transforming the Raspberry Pi into a tiny smoke machine (even though its great for parties! (this has also happened to us in the great year of 2026)).

The cable you plugged into the **5V output** goes into physical pin 4 on the pi 4. The **ground** cable goes into physical pin 34 on the pi 4. LOWKEY JUST LOOK AT A LAYOUT YOU STUPID PEANUT BRAIN JUST GO UGA BOGA I GO BACK TO MY

ROOTS OF A CAVE MAN AND DO THINGY. JUST SEARCH UP PI 4 GPIO PIN LAYOUT ITS THAT EASY. I just wanna graduate already bluh i apologize

It is to note that multiple cords can be connected to one **5V output** or **ground** unlike the power cord connections. This is useful for when connecting the fan for the pi you might need to share the connections with other wires.

3.5 Connecting the Fan

Lowkirk the fan goes into the PSU just figure out man (or woman ig), lowsmith the red is **5V** DUHHH and the black wire is the non **5V**. The ground I think it called. Idk STEVEN FIXY IT. (the red wire goes into the **5V output** and the black into the **ground** on the PSU)

4. Before Booting

Insert the SD card into the Pi and connect the Ethernet and power to the Pi

5. Logging into the Pi

Now that your Pi is booted, it's time to log into the Pi via your windows terminal.

5.1 Installing Angry IP Scanner

On your Windows machine, go to <https://angryip.org/download/> and download the installer for Windows. Once Angry IP is installed, launch it and ensure you are within the correct range for the IP. Clicking " Start Scanning " will take a few minutes to complete. Once complete, sort the results by Hostname. Search the results until you find your PI's hostname that you set earlier during the setup process. If you can't find it, then wait a few minutes and try again.

5.2 Logging into the Pi via PowerShell

Now that you have the IP of the Pi you can now log into it via Powershell, open Powershell and input the following command ``ssh (yourPi'sUsername)@(IP)`` the terminal will prompt you for the fingerprint say yes as it'll only ask this the first time, and then it'll prompt you for the pi's password input that and you should be on the pi

6. Installing Software

This section will go over everything you need to install on the Pi once you're all connected via ssh

6.1 Installing Node.js

To install node js run this command:
`sudo apt update && sudo apt install -y nodejs npm`

6.2 Installing screen

The screen command is a terminal multiplexer that allows you to run multiple terminal sessions within a single window. It is really helpful when working with Raspberry Pis. Install it using this command:

```
sudo apt install -y screen
```

Once installed, create a new named screen session for FormPix:

```
screen -S formPix
```

6.3 Cloning FormPix

Git is already installed on Raspberry Pi OS by default. If you're unfamiliar with Git, it's a version control tool used to download and manage code repositories from GitHub. Clone the FormPix repo from Smith's GitHub by running the following command: `git clone https://github.com/csmith1188/formPix/` This will download the repo to your Pi at `~/formPix`. `cd` into that directory and move onto the next step.

6.4 Using the test script

Within the FormPix directory **`npm i`** all the needed dependencies and sometimes you will need to individually install the packages since **`npm i`** doesn't work half the time in the FormPix repo. When all your packages are installed, run the following command: **`sudo node test.js`** that will run the test script to ensure your lights are configured correctly if they do not turn on after during the test **`ctrl + c`** out of the script and run **`nano test.js`** to make sure the config is correct to you pi.

6.5 Booting FormPix

After ensuring the lights work it's time to run FormPix itself, first thing is first run **`cp .env-template .env`** this copies the `.env-template` and renames it to `.env` once that's done **`nano`** into the new `.env` and edit it to match your pi's config and edit the api key to the needed key. Once that is completed run `sudo node app` and FormPix should be running success is usually determined from the output logs and if the pi is connected to speakers the boot sound should play.

8. Troubleshooting

Pi no workie? This is how fixie!

8.1 Pi Won't Boot

A variety of things could cause this. Usually, this is caused by a bad read on the SD card reader. Check this by swapping to a new Pi and inserting the SD card from that Pi into the current Pi. If it loads, then the SD card used was corrupted at some point and will need to be re-imaged. Other than that, ensure the pi is actually turned on or plugged in; you'd be surprised how much the solution was turning it on.

8.2 No Network Connection

This is usually caused by either a bad Ethernet cable or a bad port on the switch. Test the cable on another PC to ensure it works. If the cable works on the other device, then check the port on the switch by moving the cable to another port.

8.3 SSH Connection Refused

This is because you ignored the bold red and underlined text all the way back in step two... c'mon man, it's literally bolded and underlined, AND it's red, the most obvious color... How do you miss that? I can tell you one thing you aren't getting far with a mindset of that caliber here. I sure hope you didn't tell anyone your problem, they will laugh at you.

8.4 No audio playing through speakers?

If this happens, ngl, I just threw a bunch of stuff into Claude, so just do that, it'll be fine.

8.5 Anything else that I missed

Just use Google vro, not that hard.

This is the end... Lowkey didn't think graduation would come this quickly. How have we come so far? Lowkey you should listen to the bands Pierce The Veil, Hail The Sun, System Of A Down, and Hayden Parker. Thank you for coming to my Ted Talk.

Any questions about SpahrBar, go directly to the Tech lab and ask Rylan to drop everything and help you, I promise he is doing nothing important.

LOOK I KNOW YOU DID NOT READ THE ENTIRE THING. I AM GIVING YOU ONE LAST CHANCE TO READ AGAIN, BECAUSE IF IT BREAKS, IT'S ON YOU. SO ACTUALLY READ THE GOSH DARN DOCUMENT!!!!

or ask AI to make a summary...
-Rylan